

Replicating Table 2 from Luo and Spindler (2017)

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1 The Original Paper

This replication targets Luo and Spindler (2017), “L2-Boosting for Economic Applications,” published in the *American Economic Review: Papers & Proceedings* (2017, 107(5): 270–273). The paper asks whether L2-Boosting can serve as a competitive alternative to Lasso for causal inference in high-dimensional settings, where the number of potential covariates or instruments is large relative to the number of observations.

The empirical strategy builds on the orthogonal moment conditions framework of Chernozhukov et al. (2016a). Valid post-selection inference requires that the variable selection step achieve a sufficient rate of convergence. Luo and Spindler demonstrate, building on Luo and Spindler (2016), that post-L2-Boosting and orthogonal L2-Boosting satisfy these theoretical conditions and achieve the same convergence rate as Lasso. They illustrate the methods through two applications: IV estimation with many instruments using the EminentDomain dataset, and inference on a treatment effect with many controls using macroeconomic growth data.

2 The Result of Interest

This replication targets **Table 2** from the original paper, which reports the estimated effect of federal appellate court property rights decisions on log GDP per capita across four estimators. Specifically, the paper reports:

	post-Lasso	BA	post-BA	oBA
$\hat{\beta}$	0.005	0.005	0.004	0.008
s.e.	0.012	0.007	0.006	0.006

Table 2 is a natural replication target because it is the main empirical result of the paper directly answering the research question of whether L2-Boosting produces economically sensible estimates in a real high-dimensional IV setting. It is also the most self-contained result, depending only on the publicly available `EminentDomain` dataset from the `hdm` package, with no simulation infrastructure required.

3 Data and Methods

Data

Table 2 uses the `EminentDomain` dataset from the `hdm` R package (Chernozhukov et al., 2016b), originally constructed by Belloni et al. (2012). The dataset contains $n = 312$ observations with $p = 140$ potential instrumental variables capturing federal appellate court judge characteristics and court composition. The outcome is log GDP per capita (mean = 11.2297, s.d. = 0.694). The treatment variable measures the share of pro-plaintiff appellate court decisions, serving as an index of property rights protection strength.

What `preprocess.R` Does

The preprocessing script loads `EminentDomain` directly from the `hdm` package. It extracts the outcome (`y`), treatment (`d`), instruments (`z`), and controls (`x`), dropping column 50 of the control matrix following the original replication code. All variables are then standardized: the outcome is demeaned only, while the treatment, instruments, and controls are demeaned and scaled to unit variance. The processed dataset is saved to `temp/eminent_domain.rds`. The script prints a console summary confirming $n = 312$, $p = 140$, and the mean and standard deviation of log GDP.

What `analysis.R` Does

The analysis script reads `temp/eminent_domain.rds` and estimates Table 2 using four estimators: post-Lasso IV (`rlassoIV` from `hdm`), classical L2-Boosting IV (`boostSelectZ`, `post = FALSE`), post-L2-Boosting IV (`boostSelectZ`, `post = TRUE`), and orthogonal L2-Boosting IV (`orthoboostSelectZ`), the latter three from the `newboost` package. The table is written as a $\text{\LaTeX}$ fragment to `output/tables/table2.tex`. The script prints a console summary comparing replicated values to the paper’s targets.

4 Replication Results

All four estimators produce small, positive point estimates of the effect of property rights protection on log GDP per capita. The boosting-based estimators (BA, post-BA, oBA) replicate the post-Lasso sign and approximate magnitude, with modestly smaller standard errors in some specifications. The economic conclusion is unchanged for all four estimators.

Compared to the original paper, the replicated estimates across all methods match exactly when rounded to three decimal places.

5 What I Learned

The easiest part of the replication was obtaining the data. The `EminentDomain` dataset is bundled directly in the `hdm` package, requiring no download step, no data cleaning decisions, and no ambiguity about which version of the data was used. The original authors’ decision to distribute the data through a CRAN package made data extremely accessible.

The hardest part was dealing with the `newboost` package dependency. The package is not available on CRAN and must be installed from R-Forge or requested directly from the authors. R-Forge availability is intermittent and no version number or changelog is documented. I also encountered a bug in the `DGP()` function definition that causes a silent failure unless a specific global variable happens to exist in the environment. This issue was self-induced after the accidental addition of a variable in the function when running the code separately, though this emphasizes how useful Makefiles can be in replication.

A broader lesson is the value of pipeline infrastructure. The original replication package contains no Makefile, no enforced directory structure, and no README beyond a brief file description. Reproducing the paper required manually tracing dependencies across three R scripts with no automation. Building this replication with a Gentzkow–Shapiro directory structure and a Makefile made the dependency graph explicit and would allow any reader to reproduce the result with a single `make` command. If the original authors had provided even a minimal `run.all.sh`, the replication effort would have been substantially reduced.

References

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